Deep Learning CBL: Image Classification Challenge

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PS-07: Bean Leaf Lesion Classification | Date: 23 Oct 2026
GitHub: https://github.com/AathityaArun/CBL_RA2311026050026-Aathitya-A-_PS07

1. Problem and Objective

Bean farmers lose yield when angular leaf spot and bean rust go undetected. The objective is a CNN that classifies a smartphone photo of a bean leaf as angular leaf spot, bean rust or healthy, while avoiding overfitting on only 1,295 images. Success is judged by **macro F1 and per-class recall** on the held-out test set, since accuracy alone hides weak classes on a small test set.

2. Dataset and Preprocessing

Beans dataset (Makerere AI Lab, Uganda; 1,295 field images, 3 classes), loaded with the official splits: train 1,034 (345/348/341), validation 133 (44/45/44), test 128 (43/43/42). The classes are balanced, so no re-weighting was needed. The official splits are fixed and disjoint, so there is no leakage. Images were resized to 224×224 and normalised ([0,1] for the baseline; [−1,1] MobileNetV2 preprocessing). Augmentation (random flips, rotation 0.20, zoom 0.15, contrast 0.15) is applied inside the transfer model during training only, so validation and test images are never augmented. Seed = 42 (TensorFlow 2.21, GPU).

3. Methodology

Two models were compared. **Model A** is a baseline CNN from scratch (4 conv blocks, 32–256 filters, Dense-256, 13.2 M parameters) trained without augmentation or regularisation. **Model B** is MobileNetV2 (ImageNet weights) with global average pooling, Dropout 0.4, Dense-128 (L2 1e-4), Dropout 0.3 and a softmax layer. It was trained in two stages: (1) backbone frozen, head only; (2) the last 30 backbone layers unfrozen with BatchNorm kept frozen, at a lower learning rate. Early stopping (patience 5, best weights restored) and ReduceLROnPlateau (factor 0.5, min 1e-6) both monitored validation loss.

Setting	Model A (scratch)	Model B (MobileNetV2)
Loss / optimiser	Sparse CE / Adam	Sparse CE / Adam
Learning rate	1e-3	1e-3 (stage 1), 1e-4 (stage 2)
Batch size / input	32 / 224×224	32 / 224×224
Epochs	20	10 + 15
Regularisation	None	Augmentation, dropout, L2, early stopping

Table 1. Hyperparameters.

4. Results

All numbers below are from the test split (n = 128).

Model	Train acc (%)	Test acc (%)	Macro F1 (%)	Train–test gap (%)	Time (s)
A: Scratch CNN	99.81	81.25	81.52	18.56	68.09
B: MobileNetV2	96.81	93.75	93.63	3.06	80.69

Table 2. Test-set comparison (Model B per-class recall: angular leaf spot 1.00, bean rust 0.81, healthy 1.00).

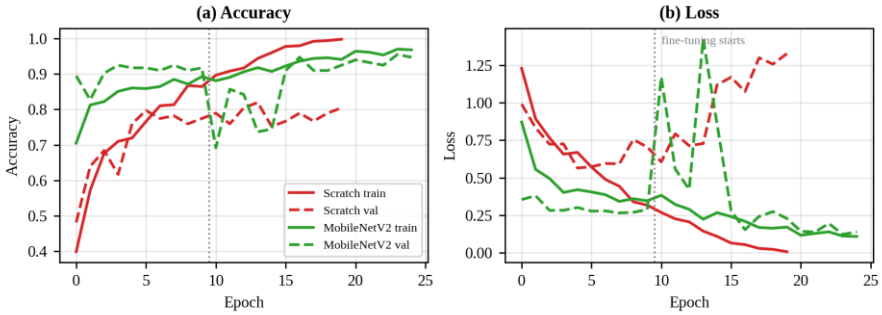


Fig. 1. Training and validation (a) accuracy and (b) loss for both models; the dotted line marks the start of fine-tuning of Model B.

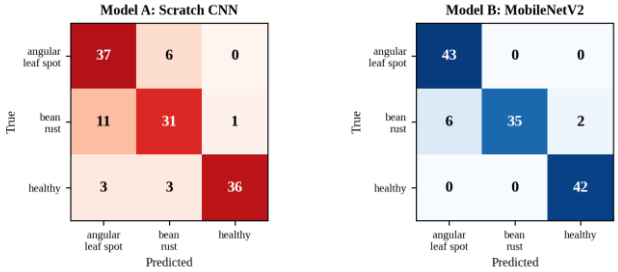


Fig. 2. Test confusion matrices: Model A (left), Model B (right).

5. Analysis and Discussion

Overfitting (the specific challenge): Model A reaches 99.81% training accuracy but only 81.25% on test, and its validation loss rises after about epoch 5 (Fig. 1), which is severe overfitting. Model B narrows the gap to 3.06 points. Its validation loss spiked briefly when fine-tuning began (lr 1e-4) and recovered within about 5 epochs. Note that Model A also lacks augmentation, so the improvement reflects transfer learning plus regularisation together, not the backbone alone. **Confusions:** all 8 errors of Model B are bean rust images (6 predicted as angular leaf spot, 2 as healthy), giving rust recall 0.81 while rust precision is 1.00 (Fig. 2). Likely causes seen in the four misclassified images: early rust pustules resemble small angular lesions; glare on waxy leaves hides fine pustules; natural yellowing or insect damage on healthy leaves; and soil or overlapping leaves in the background.

6. Real-world Relevance, Limitations and Ethics

The model could power a low-cost smartphone app for extension workers in Uganda or India for early field screening. Limitations: only two diseases, one region and one season; a 128-image test set, so a one-image change moves accuracy by 0.78 points; no testing on other phones or lighting. Missed rust (recall 0.81) could delay treatment, so predictions should support, not replace, an agronomist. Farm photos carry little privacy risk, but farmers should be told the model is not diagnostic proof.

7. Conclusion and Future Work

Fine-tuned MobileNetV2 reached 93.75% test accuracy and 93.63% macro F1, 12.50 points above the scratch baseline, while cutting the train–test gap from 18.56 to 3.06 points. Next, I would add Grad-CAM to confirm the model attends to lesions and use k-fold cross-validation to give a more stable estimate on this small dataset.

Dataset: Makerere AI Lab, "Beans Dataset", 2020, github.com/AI-Lab-Makerere/ibean; TensorFlow Datasets catalog: https://www.tensorflow.org/datasets/catalog/beans